

FORDHAM UNIVERSITY
ISGB-7905 | Web Application Development

Enda Community Runner Intelligence & Shoe Match Platform

Final Project Report

Deliverable 5

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Executive Summary

This report documents the full lifecycle of the Enda Community Runner Intelligence and Shoe Match Platform, a database-driven web application developed by Team 1 (Zaria Weekes, Misha Sundri, and Sofia Williner) for Fordham University ISGB-7905, Summer 2026.

The project was commissioned in partnership with Enda, a Kenyan performance running footwear company, and sponsored by Elis Clementino, Chief Growth Officer. The platform addresses two gaps in Enda's current operations: the absence of a personalized shoe recommendation tool, and the lack of structured data collection from the SCTY Run Club community in Brooklyn.

The platform delivers five core capabilities: runner profiling, a rule-based shoe match engine, run club event management, post-run feedback collection, and a summary reporting dashboard. It is built on a three-tier architecture using PHP, MySQL, Bootstrap 5, and Chart.js, running on Apache via XAMPP.

Requirements were gathered through an asynchronous kickoff session with Ms. Clementino, document analysis across three prior deliverables, and vendor evaluation of existing SaaS alternatives. A key insight from the sponsor shaped the project's direction: the shoe match tool is the entry point, but the platform's long-term strategic value is as a community intelligence and data asset for Enda.

The system supports 11 defined use cases across three actor types (Runner/Customer, Run Club Leader, and Enda Growth Team) and is governed by FURPS+ non-functional requirements covering usability, reliability, performance, supportability, security, and scalability. The data model consists of seven entities and is designed from the outset to support multi-club expansion beyond the SCTY Run Club pilot.

This report consolidates all prior deliverables (Vision & Scope, Kickoff Presentation, System Requirements Specification, and Analysis & Design) into a comprehensive final document, with appendices covering feasibility, data models, process models, interface design, and database design.

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1. Introduction

1.1 Organizational Background

Enda is a Kenyan performance running footwear and apparel company that designs and sells running shoes while supporting local economic development through manufacturing and community engagement initiatives. The company serves recreational and competitive runners across Kenya, the United States, and other international markets through direct-to-consumer sales and brand partnerships.

The name "Enda" derives from the Kiswahili word for "Go", a reflection of the brand's philosophy of movement, community, and forward momentum. Enda manufactures locally in Nairobi and has been expanding internationally across the Americas, Europe, and Africa.

Enda currently sponsors SCTY Run Club, a Brooklyn-based running community that operates out of Lips Cafe. Through this partnership, Enda engages directly with runners, gains visibility within the local running community, and receives informal product feedback. As the company continues to grow, there is a clear opportunity to leverage technology to better understand customer needs, improve product recommendations, and create a scalable framework for community data collection.

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1.2 Known Business Issues and Problems

Enda faced two distinct operational gaps that this project addresses:

Problem 1 — No Personalized Shoe Recommendation System:

Choosing the right running shoe is highly individual, it depends on factors such as weekly mileage, running experience, injury history, pronation tendencies, preferred terrain, and race goals. While Enda provides product information on its website, there was no mechanism for a runner to identify which Enda shoe best matched their specific needs. All product guidance was informal and unstructured.

Problem 2 — No Structured Community Data Collection:

Enda's sponsorship of SCTY Run Club provided direct access to an engaged community of runners, but no centralized system existed to collect, organize, or analyze runner feedback or participation data. Product insights were anecdotal and could not be aggregated to support business decisions such as product development, marketing, or expansion planning.

The combined effect was that Enda was unable to answer basic strategic questions: Which shoes are most popular among community runners? What surfaces do our customers typically run on? Which shoes generate the highest post-run satisfaction? This intelligence gap represented a significant missed opportunity as the company scaled.

1.3 System Objective

The objective of this project is to design, build, and deliver a web-based platform that closes both gaps simultaneously. Specifically, the system must:

- Enable runners to create a personalized profile capturing running habits, preferences, and injury history.
- Apply rule-based matching logic to recommend the most suitable Enda shoes for each runner's profile.
- Provide SCTY Run Club with event management tools, including event creation, RSVP tracking, and attendance recording.
- Allow runners to submit structured post-run feedback tied to the specific shoes they wore at each event.
- Deliver built-in summary reporting that surfaces recommendation trends, satisfaction data, and terrain distribution for Enda's growth team.

A key strategic reframe surfaced during sponsor engagement: the shoe match tool is the primary entry point for runners, but the platform's long-term value to Enda is as a community intelligence and first-party data asset. This reframe informed every design and data modeling decision made throughout the project.

1.4 Scope of the Project

The following scope was established in Deliverable 1 (Vision & Scope) and refined through sponsor feedback:

In Scope:

- Runner accounts: registration, login, and role-based access control.
- Runner profiling: intake form capturing mileage, experience, terrain, goals, injury history, pronation, and cushioning preference.
- Shoe match engine: rule-based logic scoring and ranking Enda shoes against runner profiles.
- Run club module (SCTY pilot only): event listing, RSVP/attendance tracking, and post-run feedback collection.
- Built-in summary reporting: recommendation frequency, feedback averages, and terrain distribution charts via Chart.js.
- Admin product management: Enda team can add, edit, and activate/deactivate shoe records.

Out of Scope:

- Support for multiple run clubs or community partners (SCTY pilot only; schema is extensible).
- Integration with Enda's existing e-commerce website, inventory, or payment systems.
- A separate, full administrative analytics dashboard with custom report-building.
- Ambassador program management or marketing campaign tools.
- Automated email notifications and calendar views (stretch features).

2. Requirements Gathering

2.1 Schedule and Procedure

Requirements gathering was conducted across three methods: a sponsor kickoff session, document analysis, and vendor/tool evaluation. The schedule is summarized below.

Activity	Method	Date	Participants
Kickoff Presentation Delivery	Asynchronous written exchange	June 27, 2026	Team 1 + Elis Clementino (CGO)
Sponsor Written Response Received	Written response to kickoff deck	June 27, 2026	Elis Clementino
Document Analysis (D1, D2)	Internal review of Vision & Scope + Kickoff Deck	Ongoing	Team 1
Vendor Evaluation	Evaluation of SaaS alternatives (Strava, RunSignup, Volutal)	June–July 2026	Team 1

The kickoff session was conducted asynchronously: the team prepared and delivered a 13-slide kickoff presentation to Ms. Clementino, who responded with detailed written feedback the same day. While the asynchronous format limited real-time follow-up questions, it produced more considered and specific feedback, particularly on the platform's strategic direction, than a live Q&A might have.

Vendor evaluation assessed four categories: shoe recommendation SaaS platforms, run club management tools, development stacks, and reporting/analytics tools. In each category, existing solutions were found to be either too narrowly scoped, hardware-dependent, or part of closed ecosystems that could not be embedded into a custom application.

2.2 Results and Key Findings

The requirements gathering process produced five major insights that shaped the final specification:

Finding 1 — Strategic Reframe: Data over Recommendations

Ms. Clementino clarified that the shoe match tool, while valuable, was not the platform's most important feature. The real long-term asset is the data, specifically, the platform's ability to answer community intelligence questions: What surfaces do our customers run on? Which shoes produce the highest satisfaction? How do beginner runners differ from marathon runners? This reframe shifted the project's emphasis toward data model extensibility, aggregation logic, and reporting depth.

Sponsor quote: "The real long-term asset is the data." — Elis Clementino, CGO, Enda

Finding 2 — Scalability as a Design Principle

The sponsor asked that even within a single-club pilot, the platform's architecture be designed to scale to other run clubs, ambassadors, retailers, university teams, race events, and community partners. This informed the decision to introduce a run_clubs entity in the data model rather than hard-coding SCTY-specific values.

Finding 3 — Custom Build Required

Vendor evaluation confirmed that no existing SaaS tool addressed all three requirements (runner profiling, shoe matching, and run club management) within a single platform. A fully custom build using the course-required PHP/MySQL stack was the only viable approach.

Finding 4 — Feedback Tied to Specific Shoes

Post-run feedback should capture which specific Enda shoe the runner wore during the event. This allows the platform to produce per-shoe satisfaction averages — a key output for Enda's product team. This insight was incorporated into UC-05.

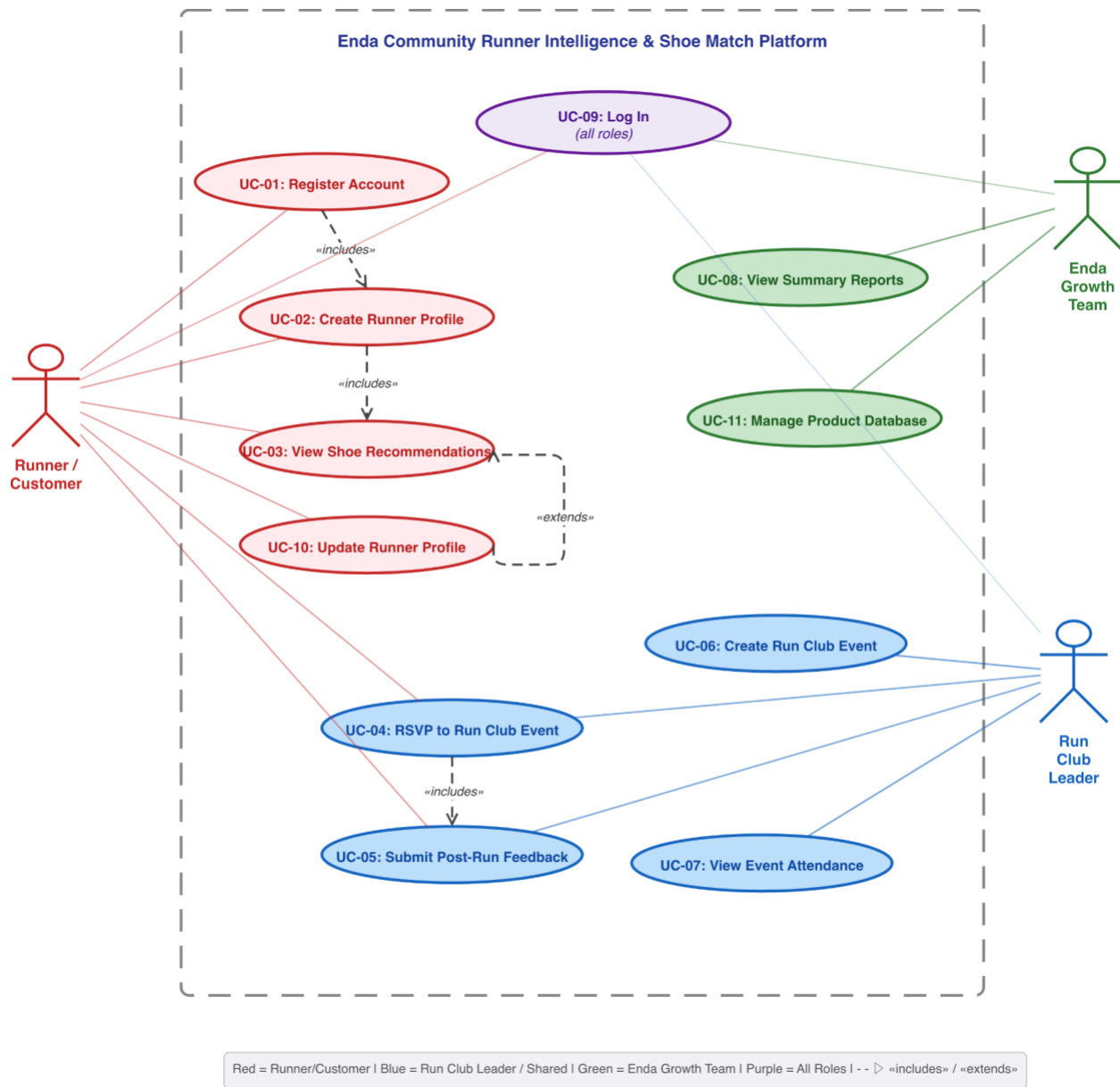
Finding 5 — "Why Now" Context

Ms. Clementino highlighted broader market context: community commerce is growing, first-party customer data is increasingly valuable as third-party tracking shrinks, and run clubs have become one of the strongest customer acquisition channels in the running industry. This context reinforces the business case for the platform and was incorporated into the project narrative.

3. Functional Requirements: Use Cases

3.1 Use Case Diagram Description

The system boundary for the Enda Community Runner Intelligence and Shoe Match Platform encompasses 11 use cases, three primary actors, and four include/extend relationships.



Three primary actors interact with the system:

- Runner/Customer — the end user who creates a profile, views shoe recommendations, RSVPs to events, and submits post-run feedback.
- Run Club Leader — manages event creation and attendance tracking for SCTY Run Club.
- Enda Growth Team — views summary reports and manages the shoe product database.

Include/Extend relationships:

- UC-01 (Register Account) includes UC-02 (Create Runner Profile) — a profile requires an account.
- UC-03 (View Shoe Recommendations) includes UC-02 — a complete profile is required to generate matches.
- UC-10 (Update Runner Profile) includes UC-03 — updating the profile re-triggers the match engine.
- UC-05 (Submit Post-Run Feedback) includes UC-04 — a runner must have an RSVP record before feedback can be submitted.

Use Case ID	Name	Primary Actor	D4 Status
UC-01	Register Account	Runner/Customer	Carried forward, no change
UC-02	Create Runner Profile	Runner/Customer	Carried forward, no change
UC-03	View Shoe Recommendations	Runner/Customer	Updated: match rationale labels added
UC-04	RSVP to Run Club Event	Runner/Customer	Carried forward, no change
UC-05	Submit Post-Run Feedback	Runner/Customer	Updated: shoe worn field added
UC-06	Create Run Club Event	Run Club Leader	Carried forward, no change
UC-07	View Event Attendance	Run Club Leader	Carried forward, no change
UC-08	View Summary Reports	Enda Growth Team	Updated: terrain distribution chart added
UC-09	Log In	All user types	Carried forward, no change
UC-10	Update Runner Profile	Runner/Customer	Carried forward, no change
UC-11	Manage Product Database	Enda Growth Team	Updated: is_active toggle added

3.2 Detailed Use Case Descriptions

UC-01: Register Account

Actor: Runner/Customer

Description: A new user creates an account by providing their name, email address, and password. The system validates the input, checks for duplicate emails, and creates the account.

Precondition: User has not previously registered.

Postcondition: Account is created and user is redirected to the profile intake form.

Main Flow:

1. User navigates to the registration page.
2. User enters name, email, and password.
3. System validates all fields and checks for duplicate email.
4. System creates account and initiates a session.
5. System redirects user to the runner profile intake form (UC-02).

Alternative/Exception: If email already exists, system displays an error and prompts the user to log in instead.

UC-02: Create Runner Profile

Actor: Runner/Customer

Description: A registered user completes the runner intake form capturing weekly mileage, experience level, preferred terrain, race goals, injury history, and pronation/cushioning preferences. This data drives the shoe match engine.

Precondition: User is logged in and has not yet completed a profile.

Postcondition: Profile is stored in the database and shoe match results are generated.

Main Flow:

- 1. User is presented with the runner intake form.
- 2. User selects or inputs: weekly mileage, experience level, terrain preference, race goals, injury history, pronation type, and cushioning preference.
- 3. User submits the form.
- 4. System validates all required fields.
- 5. System saves the profile and triggers the shoe match logic.
- 6. System redirects user to their shoe recommendations page (UC-03).

Alternative/Exception: If required fields are missing, system highlights them and prevents submission.

UC-03: View Shoe Recommendations

Actor: Runner/Customer

Description: The system presents a ranked list of Enda shoes based on the runner's profile attributes. Each recommendation displays shoe name, key attributes, and a match rationale label (e.g., "Matched on: road terrain, moderate cushioning").

Precondition: User is logged in and has a complete runner profile.

Postcondition: User views personalized shoe recommendations.

Main Flow:

- 1. User navigates to the shoe match results page.
- 2. System queries the Enda product database using the runner's profile attributes.
- 3. System applies matching logic and ranks shoes by compatibility.
- 4. System displays ranked shoe list with name, attributes, and match reasoning.
- 5. User can click on a shoe for more detail.

Alternative/Exception: If no close match is found, system displays the closest available options with a note.

UC-04: RSVP to Run Club Event

Actor: Runner/Customer

Description: A runner views the SCTY Run Club event listing and RSVPs to an upcoming event. The system records their attendance intent.

Precondition: User is logged in. At least one upcoming event exists.

Postcondition: RSVP record is created and confirmed to the user.

Main Flow:

- 1. User navigates to the events listing page.
- 2. User selects an upcoming event and clicks the RSVP button.
- 3. System creates an attendance record linking user and event.
- 4. System displays a confirmation message inline (no full page reload).

Alternative/Exception: If user has already RSVPed, system shows their existing status with an option to cancel.

UC-05: Submit Post-Run Feedback

Actor: Runner/Customer

Description: After an event, a runner completes a short feedback form rating shoe comfort, performance, and durability. The form includes a field capturing which specific Enda shoe they wore.

Precondition: User has an RSVP record for the event. Event has taken place.

Postcondition: Feedback is stored and factored into summary reports.

Main Flow:

- 1. User navigates to past events and opens the feedback form.
- 2. User selects which shoe they wore (or "Other / Non-Enda").
- 3. User rates comfort, performance, and durability (1–5 stars each).
- 4. User optionally adds a written comment.
- 5. System saves the feedback record and updates aggregate views.

Alternative/Exception: Duplicate submission attempt shows existing response; edit is stretch scope for the pilot.

UC-06: Create Run Club Event

Actor: Run Club Leader

Description: A club leader creates a new event for SCTY Run Club, specifying name, date, time, location, and optional notes.

Precondition: User is logged in with the "leader" or "admin" role.

Postcondition: Event is listed on the platform and open for RSVPs.

Main Flow:

- 1. Leader navigates to the event management screen.
- 2. Leader clicks "Create New Event" and fills in the form.
- 3. System validates required fields and saves the event record.
- 4. Event appears in the public events listing.

Alternative/Exception: Missing required fields prevent submission; inline validation highlights errors.

UC-07: View Event Attendance

Actor: Run Club Leader

Description: A leader views the RSVP roster for any event and can update attendance status (RSVPed / Attended / Absent) for each runner.

Precondition: User is logged in as a leader or admin. The event exists.

Postcondition: Attendance statuses are saved and visible in the event roster.

Main Flow:

- 1. Leader navigates to event management and clicks "View Roster."

- 2. System displays a table of RSVPed runners with attendance status dropdowns.
- 3. Leader updates statuses and clicks "Save Attendance."
- 4. System performs a batch update of all RSVP records.

UC-08: View Summary Reports

Actor: Enda Growth Team

Description: An Enda admin views built-in reporting visualizations including most-recommended shoes, average feedback ratings by shoe, and terrain distribution across runner profiles. Rendered via Chart.js.

Precondition: User is logged in with the "admin" role.

Postcondition: Report data is displayed; no write occurs.

Main Flow:

- 1. Admin navigates to the Summary Reports screen.
- 2. System queries across runner_profiles, shoes, rsvps, and feedback tables.
- 3. System renders three Chart.js visualizations: recommendation frequency bar chart, feedback averages grouped bar chart, and terrain distribution donut chart.
- 4. Summary stats (total registered runners, events held, feedback submissions) appear above charts.

Performance requirement: reports must load within 5 seconds at expected pilot data volumes.

UC-09: Log In

Actor: All user types (Runner, Leader, Admin)

Description: A registered user authenticates by providing their email and password.

Precondition: User has a registered account.

Postcondition: Session is created; user is redirected to their role-appropriate dashboard.

Main Flow:

- 1. User navigates to the login page.
- 2. User enters email and password.
- 3. System validates credentials against bcrypt hash.
- 4. System creates a PHP session storing user_id and role.
- 5. System redirects to dashboard based on role.

Alternative/Exception: After three failed attempts, account is temporarily locked.

UC-10: Update Runner Profile

Actor: Runner/Customer

Description: A runner returns to update their profile — for example, after increasing mileage or recovering from an injury. The updated profile re-triggers the shoe match engine.

Precondition: User is logged in and has an existing profile.

Postcondition: Profile is updated and new shoe recommendations are generated.

Main Flow:

- 1. User navigates to their profile page.
- 2. User edits one or more profile fields and submits the updated form.
- 3. System validates and saves the changes.
- 4. System re-runs shoe match logic and redirects to updated recommendations.

Alternative/Exception: If required fields are cleared, system prevents submission and highlights errors.

UC-11: Manage Product Database

Actor: Enda Growth Team

Description: An Enda admin adds or updates shoe records in the product database, including attributes such as cushioning level, stability, flexibility, weight, terrain suitability, and active/inactive status.

Precondition: Admin is logged in with the "admin" role.

Postcondition: Product database is updated and shoe match logic reflects new data.

Main Flow:

- 1. Admin navigates to product management.
- 2. Admin selects an existing shoe to edit or clicks "Add New Shoe."
- 3. Admin enters or updates shoe attributes including the is_active toggle.
- 4. System updates the database. Inactive shoes are excluded from future match results but remain linked to historical feedback records.

Alternative/Exception: Duplicate shoe name prompts admin to confirm or rename.

4. Non-Functional Requirements

Non-functional requirements define the quality attributes, constraints, and operational characteristics the system must satisfy. These are organized using the URPS+ categories of the FURPS+ framework.

Usability (U)

- The platform must be usable by runners with no prior technical training. The profile form, shoe match results, and RSVP flow must be completable by a first-time user without any guidance or tutorial.
- All primary user actions: registering, completing a profile, viewing shoe recommendations, and RSVPing must be reachable within three clicks from the home screen.
- The interface must be responsive and functional on both desktop and mobile browsers, as runners are likely to access the platform on their phones before or after a run.
- Error messages must be written in plain language and must clearly indicate what the user needs to correct.
- Forms must provide inline validation feedback before submission rather than returning the user to an empty form after an error.

Reliability (R)

- The platform must be available during expected usage windows, including evenings and weekends when run club events typically occur.
- Runner profile data and post-run feedback must not be lost in the event of a page reload or brief session interruption. Draft form data should persist where technically feasible.
- Database write operations (profile saves, RSVP records, feedback submissions) must be atomic, either fully committed or fully rolled back, to prevent partial or corrupt records.

Performance (P)

- Shoe match results must be returned and rendered within three seconds of a runner submitting or updating their profile, under normal load conditions.
- The events listing page must load within two seconds.
- The summary reporting screen must aggregate and display results within five seconds, given expected pilot data volumes.

Supportability (S)

- The codebase must follow consistent naming conventions and include inline comments sufficient for any team member to understand and modify any module.
- The database schema must be documented separately (ERD) so future developers can extend the data model without reverse-engineering the application code.
- The shoe product database must be editable by an authorized Enda team member through the admin interface, without requiring code changes or developer involvement.
- The run club module must be implemented as a modular component so that support for additional clubs can be added in a future release without rewriting the core application.

+ Additional Constraints

Technology:

The platform is built using PHP 8.x and MySQL 8.x as the server-side and database technologies, consistent with course requirements. Front-end implementation uses HTML, CSS (Bootstrap 5), and JavaScript (Chart.js for reporting visualizations). Apache is provided via XAMPP for local development.

Security:

User passwords are stored as bcrypt-hashed values (PHP password_hash()). The application protects against SQL injection through PDO prepared statements. Session tokens expire after 30 minutes of inactivity. Role-based access control (RBAC) is enforced at the PHP layer on every protected page.

Data Privacy:

Runner profile data — including injury history and personal running details — is stored securely and must not be exposed in client-side code or API responses to unauthorized users.

Scalability:

Although the pilot is limited to SCTY Run Club, the database schema is designed to support multiple clubs without structural changes. Run club-specific data is associated with a run_clubs entity, not hard-coded.

Scope:

The platform does not integrate with Enda's existing e-commerce website, internal inventory systems, or any third-party payment processor during this semester. Shoe purchase links reference the Enda website externally.

5. Plan for Next Steps

5.1 Unfinished Work for Design and Development

- Source code finalization and deployment verification: the production domain was pending final deployment at the time of this submission. The GitHub repository contains the full codebase; hosting on a shared PHP/MySQL server is the next step.
- Automated email notifications: identified as a stretch feature. Post-RSVP and post-event feedback prompt emails were not implemented in the pilot.
- Calendar view: the events listing was built as a simple list view per scope. A calendar UI is deferred to a future iteration.
- Feedback edit functionality: runners can view but not edit existing feedback submissions in the pilot. Edit capability was noted as stretch scope.
- Mobile testing: the interface is built on Bootstrap 5 responsive grid, but formal mobile device testing across iOS and Android browsers is recommended before a public launch.

5.2 Plan for Migration, Conversion, and Deployment

- The application was developed locally using XAMPP (Apache + MySQL). Deployment to production requires a shared hosting environment with PHP 8.x and MySQL 8.x support (e.g., Hostinger, SiteGround, or Enda's preferred provider).
- The MySQL schema and seed data (shoe product records) can be exported as a .sql dump and imported on the production server.
- The GitHub repository serves as the source of truth; deployment involves copying the codebase to the server's web root and updating database connection strings in the PHP config file.
- No data migration from a legacy system is required, as this is a new platform.

5.3 Training and Support

- Three admin roles require onboarding: Enda Growth Team (product management and reporting), Run Club Leaders (event creation and attendance), and general runners (profile creation is self-guided).
- A brief admin guide documenting the product management and reporting screens is recommended for handoff to Enda.
- The shoe product database must be seeded with accurate Enda product data by the Enda Growth Team prior to public launch.

5.4 Longer-Term Product Roadmap

Based on sponsor feedback, the following capabilities represent high-value future additions beyond the current semester scope:

- Multi-club support: onboard additional run clubs, ambassadors, retailers, university teams, and race events as the platform scales.
- Training logs and milestones: allow runners to log activity, track mileage over time, and earn milestone achievements.
- Community challenges: group running goals tied to run club events and shoe performance.
- Advanced analytics: expand the summary reporting dashboard with trend analysis, cohort comparisons, and export capabilities for Enda's product and marketing teams.
- Machine learning: replace the rule-based match engine with a data-driven model as runner and feedback data accumulates.

6. Postmortem Reflections

The following reflections document what the team learned over the course of the project — what worked, what presented unexpected challenges, and what we would do differently if starting again.

What Went Well

- The sponsor's written feedback at kickoff was unusually substantive and gave the team a clear strategic direction early. The reframe from "shoe recommendation engine" to "community intelligence platform" sharpened every subsequent design decision.
- The decision to pilot with a single run club (SCTY) while designing the data model for multi-club extensibility proved to be the right balance between manageable scope and future-proof architecture.
- The FURPS+ framework provided an effective structure for capturing both functional and non-functional requirements, and made it easier to trace design decisions back to stated requirements throughout the project.
- Dividing the use cases across team members for development while maintaining a shared database schema kept parallel work organized.

Challenges Encountered

Sponsor Availability

Coordinating with Ms. Clementino presented consistent scheduling challenges. As Chief Growth Officer of an international company, her calendar was less flexible than anticipated, and several planned follow-up sessions were difficult to schedule. This meant that some requirements — particularly around feedback form design and report prioritization — were finalized without the additional sponsor validation the team had initially planned. Going forward, we would establish a more structured communication cadence from day one: a recurring check-in, even briefly, every two weeks.

Access to Internal Product Data

Enda's internal product catalog was not accessible due to data availability constraints. The shoe database was seeded using publicly available product specifications from Enda's website. While functionally sufficient for the pilot, this limited the accuracy of the shoe match engine for models with less publicly documented technical attributes.

What We Would Do Differently

Start Coding Earlier

A significant lesson from this project is the importance of beginning implementation as soon as the core data model is stable — not waiting for all requirements documentation to be finalized. In retrospect, the team spent more time in the requirements and modeling phases than necessary before beginning the build. Earlier coding would have surfaced integration issues and edge cases sooner, leaving more time for testing and refinement before the final deadline.

Schedule More Structured Sponsor Check-Ins

Rather than relying on ad hoc communication, the team would build a formal sponsor meeting schedule into the project plan from the outset — including sending consolidated question lists before each meeting to make the most of limited sponsor time.

Parallel Development Earlier

The team's workflow tended toward sequential phases (requirements → design → build) rather than overlapping them. Earlier parallelization — beginning front-end wireframes during the requirements phase, or setting up the database schema as soon as the ERD was drafted — would have compressed the timeline.

Appendix A: Feasibility and Cost-Benefit Analysis

Technical Feasibility

The platform is built on the PHP/MySQL/XAMPP stack required by the course. All team members had prior experience with this stack, reducing ramp-up time. Bootstrap 5 and Chart.js are well-documented open-source libraries that handle responsive layout and reporting visualization without additional dependencies. No proprietary APIs, hardware integrations, or machine learning components are included in the pilot, keeping the technical risk low.

Operational Feasibility

The platform is designed for self-service use by runners and club leaders without technical training. The run club module requires only minimal admin involvement (event creation and attendance marking). Enda's team can manage the shoe product database through the admin interface without developer involvement.

Schedule Feasibility

By narrowing scope to a single run club and replacing the full analytics dashboard with built-in Chart.js views, the two core features (shoe matching and run club management) were individually achievable within the summer semester timeline. The primary schedule risk was sponsor availability rather than build complexity.

Cost-Benefit Analysis

Development costs are limited to team time (student project) and hosting costs.

Cost Item	Estimated Cost	Notes
Development (student labor)	\$0 (academic project)	3 team members across ~14 weeks
Hosting (post-semester)	~\$5–15/month	Shared PHP/MySQL hosting (Hostinger, SiteGround, etc.)
Domain name (optional)	~\$10–15/year	If Enda opts to deploy publicly
Open-source tools (PHP, MySQL, Bootstrap, Chart.js)	\$0	All MIT or compatible licenses

Primary benefits:

- First-party runner data asset: enables Enda to answer community intelligence questions that were previously unanswerable.
- Improved conversion: runners who receive a personalized shoe match are more likely to purchase the recommended product.
- Scalable infrastructure: the schema and architecture can support additional run clubs, ambassadors, and community partners with minimal additional development.
- Competitive differentiation: few performance shoe brands at Enda's scale have a dedicated community intelligence platform.

Appendix B: Data Dictionary

The following tables define each entity in the MySQL database schema, including all attributes, data types, constraints, and descriptions.

Entity 1: users

Stores all platform user accounts. Role differentiates runners, run club leaders, and Enda admins.

Attribute	Data Type	Constraints	Description
user_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique user identifier
name	VARCHAR(100)	NOT NULL	Full display name
email	VARCHAR(150)	UNIQUE, NOT NULL	Login email address
password_hash	VARCHAR(255)	NOT NULL	bcrypt-hashed password
role	ENUM	NOT NULL, DEFAULT 'runner'	Values: 'runner', 'leader', 'admin'
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

Entity 2: runner_profiles

Stores the intake form data for each runner. Primary input to the shoe match engine.

Attribute	Data Type	Constraints	Description
profile_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique profile identifier
user_id	INT	FK → users.user_id, UNIQUE, NOT NULL	One profile per user
weekly_mileage	ENUM	NOT NULL	Values: 'under_10', '10_to_20', '20_to_40', '40_plus'
experience_level	ENUM	NOT NULL	Values: 'beginner', 'intermediate', 'advanced', 'elite'
terrain_preference	ENUM	NOT NULL	Values: 'road', 'trail', 'track', 'mixed'
race_goals	ENUM	NOT NULL	Values: 'casual', '5k_10k', 'half_marathon', 'full_marathon', 'ultra'
injury_history	TEXT	NULL	Free-text field; optional
pronation_type	ENUM	NOT NULL	Values: 'neutral', 'overpronation', 'supination', 'unknown'
cushioning_preference	ENUM	NOT NULL	Values: 'minimal', 'moderate', 'maximum'

Attribute	Data Type	Constraints	Description
updated_at	DATETIME	NOT NULL	Timestamp of last profile update

Entity 3: shoes

Stores the Enda product catalog. Attributes are directly mapped to runner profile attributes for matching.

Attribute	Data Type	Constraints	Description
shoe_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique shoe identifier
name	VARCHAR(150)	NOT NULL	Shoe model name (e.g., 'Iten', 'Lapatet')
cushioning_level	ENUM	NOT NULL	Values: 'minimal', 'moderate', 'maximum'
stability_rating	TINYINT(1)	NOT NULL	Scale 1–5; higher = more stable
flexibility_rating	TINYINT(1)	NOT NULL	Scale 1–5; higher = more flexible
weight_grams	INT	NOT NULL	Shoe weight in grams (single shoe)
terrain_suitability	ENUM	NOT NULL	Values: 'road', 'trail', 'track', 'mixed'
description	TEXT	NULL	Marketing/product description
enda_url	VARCHAR(255)	NULL	External link to Enda product page
is_active	TINYINT(1)	NOT NULL, DEFAULT 1	1 = active in matching; 0 = retired

Entity 4: run_clubs

Stores run club records. Introduced to ensure the data model supports multi-club scalability per sponsor request.

Attribute	Data Type	Constraints	Description
club_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique club identifier
name	VARCHAR(150)	NOT NULL	Club name (e.g., 'SCTY Run Club')
location	VARCHAR(255)	NULL	Primary meeting location
description	TEXT	NULL	Club description
is_active	TINYINT(1)	NOT NULL, DEFAULT 1	Whether the club is currently active

Entity 5: events

Stores run club events created by leaders.

Attribute	Data Type	Constraints	Description
event_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique event identifier
club_id	INT	FK → run_clubs.club_id, NOT NULL	The run club hosting this event
created_by	INT	FK → users.user_id, NOT NULL	The leader who created the event
event_name	VARCHAR(200)	NOT NULL	Display name for the event
event_date	DATE	NOT NULL	Date of the event
event_time	TIME	NULL	Start time (optional)
location	VARCHAR(255)	NOT NULL	Event location / meeting point
notes	TEXT	NULL	Optional leader notes
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

Entity 6: RSVPS

Junction table resolving the many-to-many relationship between users and events. Also tracks attendance status.

Attribute	Data Type	Constraints	Description
rsvp_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique RSVP record identifier
event_id	INT	FK → events.event_id, NOT NULL	The event being RSVPed to
user_id	INT	FK → users.user_id, NOT NULL	The runner RSVPing
rsvp_date	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the RSVP was made
attendance_status	ENUM	NOT NULL, DEFAULT 'rsvped'	Values: 'rsvped', 'attended', 'absent'

Unique constraint: (event_id, user_id) — prevents duplicate RSVPs.

Entity 7: feedback

Stores post-run feedback submitted by runners after events.

Attribute	Data Type	Constraints	Description
feedback_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique feedback record identifier
event_id	INT	FK → events.event_id, NOT NULL	The event the feedback is for
user_id	INT	FK → users.user_id, NOT NULL	The runner submitting feedback
shoe_id	INT	FK → shoes.shoe_id, NULL	The Enda shoe worn (nullable)
comfort_rating	TINYINT(1)	NOT NULL	Scale 1–5
performance_rating	TINYINT(1)	NOT NULL	Scale 1–5
durability_rating	TINYINT(1)	NOT NULL	Scale 1–5
comment	TEXT	NULL	Optional written comment
submitted_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Submission timestamp

Unique constraint: (event_id, user_id) — prevents a runner from submitting feedback twice for the same event.

Appendix C: Architecture Design

Three-Tier Architecture Overview

The platform follows a standard three-tier client-server architecture:

Presentation Tier:

HTML/CSS rendered via Bootstrap 5, with JavaScript for client-side validation and Chart.js for report visualizations. Pages are server-rendered PHP templates; no separate front-end framework is used.

Application Tier:

PHP (server-side logic, session management, shoe match engine, and routing). All form processing, authentication, and database interaction occur here. PDO prepared statements are used throughout to prevent SQL injection.

Data Tier:

MySQL relational database accessed via PHP PDO. XAMPP (Apache + MySQL) is used as the local development environment.

Request Flow

A user action in the browser (e.g., submitting the profile form) sends an HTTP POST to an Apache-served PHP file. The PHP controller validates input, executes parameterized SQL queries against the MySQL database, and returns a rendered HTML response or redirect. No AJAX framework is used; full-page responses are standard, with lightweight JavaScript used only for inline form validation and RSVP confirmation messages.

Authentication and Session Management

User sessions are managed via PHP native sessions (\$_SESSION). Upon successful login, the user's user_id and role are stored in the session. Role-based access control (RBAC) is enforced at the top of each PHP page by checking the session role before rendering any content. Sessions expire after 30 minutes of inactivity.

Hosting and Deployment

During development and the course pilot, the application runs locally via XAMPP on each team member's machine, with a shared MySQL dump used for synchronization. The GitHub repository serves as the single source of truth. For post-semester deployment, a standard shared hosting environment with PHP/MySQL support (e.g., Hostinger, SiteGround) would be sufficient.

Technology Stack Summary

Tool	Role	License
PHP 8.x	Server-side application logic	Open source (PHP License)
MySQL 8.x	Relational database	Open source (GPL)
Apache (via XAMPP)	Local web server	Open source (Apache License)
Bootstrap 5	Responsive front-end CSS framework	Open source (MIT)
Chart.js	JavaScript charting for summary reports	Open source (MIT)
GitHub	Version control and team collaboration	Free tier (SaaS)

Appendix D: Page Map and User Interface Inventory

Page Map

The platform's pages are organized across three user-facing contexts: the public layer, the runner dashboard, and the admin/leader layer.

[Home / Landing Page]

└── [Register] → [Runner Profile Form] → [Shoe Recommendations]

└── [Login]

└── Runner View

| └── [Runner Dashboard]

| | └── [My Profile] → [Edit Profile] → [Shoe Recommendations]

| | └── [Shoe Recommendations] → [Shoe Detail]

| | └── [Events Listing] → [Event Detail + RSVP]

| | └── [My Past Events] → [Feedback Form]

└── Run Club Leader View

| └── [Leader Dashboard] → [Event Management] → [Create Event] /

[Event Roster]

└── Enda Admin View

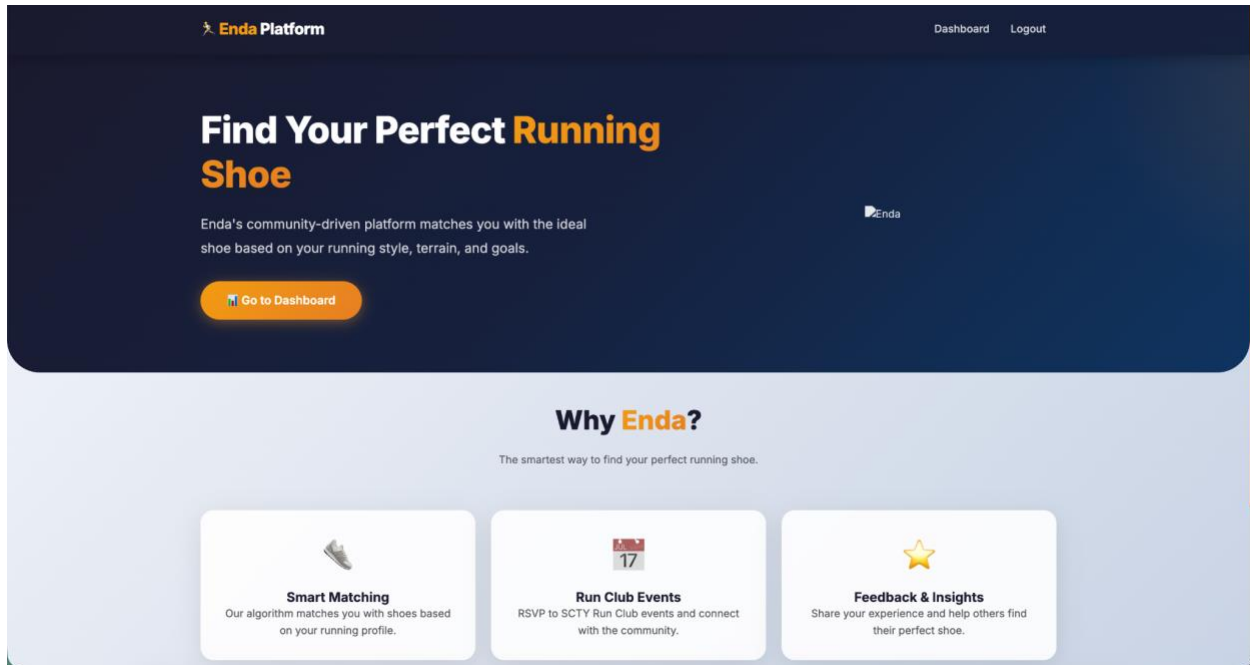
└── [Admin Dashboard] → [Product Management] / [Summary Reports]

UI Inventory

Page / Screen	Primary Actor	Linked Use Case(s)
Home / Landing Page	Public	None (navigation only)
Register	Runner/Customer	UC-01
Log In	All users	UC-09
Runner Profile Form	Runner/Customer	UC-02
Shoe Recommendations	Runner/Customer	UC-03
Shoe Detail Page	Runner/Customer	UC-03
Events Listing	Runner/Customer	UC-04
Event Detail + RSVP	Runner/Customer	UC-04
Feedback Form	Runner/Customer	UC-05
Runner Dashboard	Runner/Customer	UC-03, UC-04
Edit Profile	Runner/Customer	UC-10
Leader Dashboard / Event Management	Run Club Leader	UC-06, UC-07
Create/Edit Event Form	Run Club Leader	UC-06
Event Roster	Run Club Leader	UC-07
Product Management	Enda Growth Team	UC-11

Page / Screen	Primary Actor	Linked Use Case(s)
Add/Edit Shoe Form	Enda Growth Team	UC-11
Summary Reports	Enda Growth Team	UC-08

UI Screenshots:



Join Enda

Create your free account.

Full Name

Email Address

Password

Confirm Password

Create Account

Already have an account? [Log In](#)

Welcome Back

Log in to your Enda account.

Email Address

Password

Log In

Don't have an account? [Register](#)

Your Runner Profile

Tell us about yourself so we can recommend the perfect shoe.

Weekly Mileage

20-40 mi



Experience Level

Elite



Preferred Terrain

Mixed



Race Goals

Ultra



Pronation Type

Neutral



Cushioning Preference

Minimal / Lightweight



Injury History (optional)

e.g., plantar fasciitis, IT band issues



Save Profile & See My Shoes

Your Perfect Match

Based on your running profile, here are your top recommendations.

Your Profile

Mixed Terrain Minimal Cushioning Neutral Pronation
20-to-40 Mileage ultra

Tai Jogger

#1 Match

Match Score 80%

Mixed Moderate

Stability Flexibility Weight
3/5 4/5 240g

✓ Why this shoe:

- Terrain: Mixed (exact match)
- Cushioning: Moderate (close match)
- Flexibility: 4/5 (good for race distance)

 [View on Enda.co →](#)

Iten

#2 Match

Match Score 55%

Road Moderate

Stability Flexibility Weight
4/5 4/5 220g

✓ Why this shoe:

- Terrain: Road (good for mixed)
- Cushioning: Moderate (close match)
- Flexibility: 4/5 (good for race distance)

 [View on Enda.co →](#)

Koobi Fora

#3 Match

Match Score 45%

Trail Moderate

Stability Flexibility Weight
5/5 3/5 280g

✓ Why this shoe:

- Terrain: Trail (good for mixed)
- Cushioning: Moderate (close match)

 [View on Enda.co →](#)

[Update Profile](#)

[Back to Dashboard](#)

Enda Platform

Logout

Welcome back, Misha Sundri 🙌

Your running community dashboard.

🏆 Your Top Shoe Match

Complete your profile to get shoe recommendations.

Complete Profile

👤 My Profile

👟 Shoe Matches

📅 17 Events

★ Feedback

📊 Your Stats

0 Events

0 Feedback

0 RSVPs

📅 17 Run Club Events

Back to Dashboard

Upcoming Events

No upcoming events. Check back later!

Past Events

No past events yet.

← → ↺ ⚠ Not Secure endacommunity.site.je/create_event.php

Access denied. You must be a leader or admin.

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Appendix E: Database Reports and Queries

The following reports and queries are implemented within the platform's Summary Reports screen (UC-08) and serve as the primary analytical outputs for the Enda Growth Team.

Report / Query	Description	Data Sources	Visualization
Most-Recommended Shoes	Counts how many runner profiles each shoe has been matched to, ranked descending.	runner_profiles, shoes (via match engine logic)	Horizontal bar chart (Chart.js)
Average Feedback Ratings by Shoe	Calculates average comfort, performance, and durability ratings per shoe model.	feedback, shoes	Grouped bar chart (Chart.js)
Runner Terrain Distribution	Shows the breakdown of terrain_preference across all runner profiles (road, trail, track, mixed).	runner_profiles	Pie / donut chart (Chart.js)
Summary Stats (header)	Displays total registered runners, total events held, and total feedback submissions.	users, events, feedback	Stat cards (HTML)
Event Roster Export (leader view)	Lists all RSVPs for a given event with attendance status. Used by run club leaders for event management.	rsvps, users, events	HTML table (UC-07 screen)

Appendix F: Data Models & Process Models

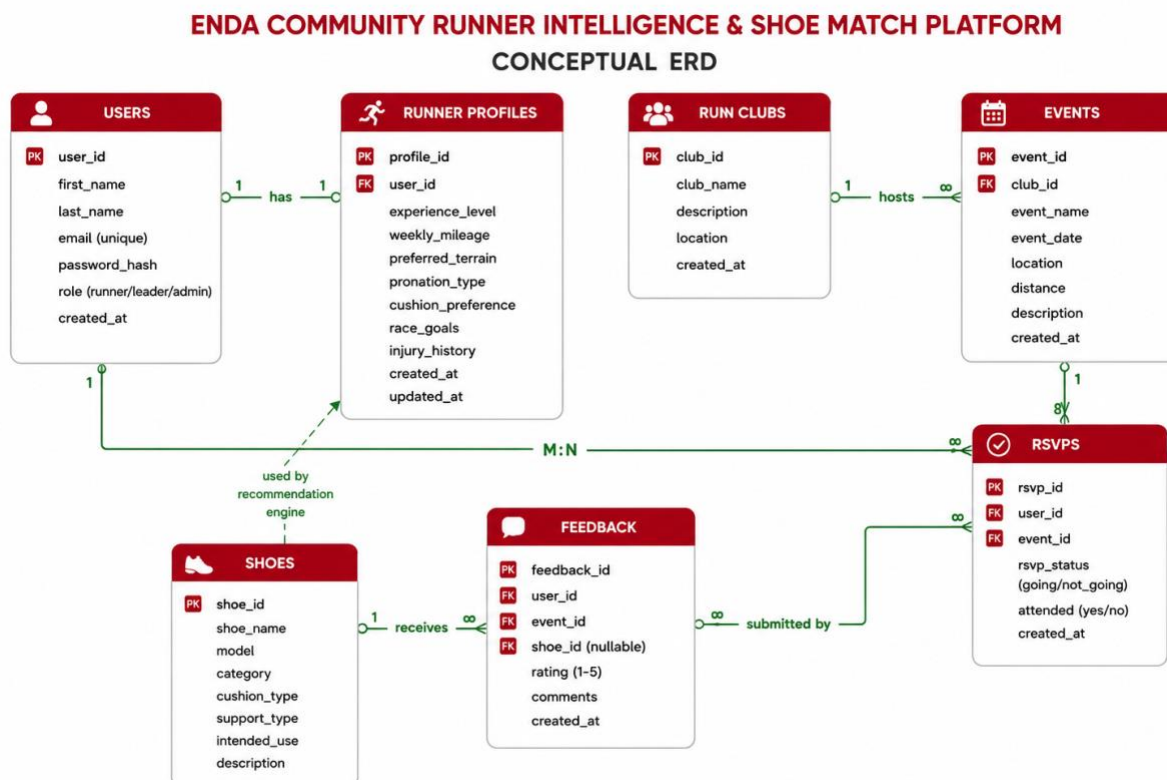
This appendix presents the data model and all process models developed during the design phase of the project. Each section includes a written description followed by the visual diagram.

Entity Relationship Diagram (ERD)

The database schema consists of seven entities. The users table is the central entity — it stores all platform accounts and differentiates roles (runner, leader, admin) through a role attribute. Each user has exactly one runner_profile, which captures the intake form data used as input to the shoe match engine. The shoes table stores the Enda product catalog; its attributes (cushioning level, terrain suitability, pronation compatibility) are structurally mirrored against runner profile attributes to enable rule-based matching.

The run_clubs table stores club records and was introduced to support multi-club extensibility, even though the pilot runs with a single club (SCTY). Each run club hosts one or more events, created by a user with the leader role. The rsvps table is a junction entity resolving the many-to-many relationship between users and events, and also carries the attendance status field (RSVPed / Attended / Absent). The feedback table stores post-run ratings submitted by runners after an event; each feedback record is linked to a specific event, user, and optionally a specific shoe.

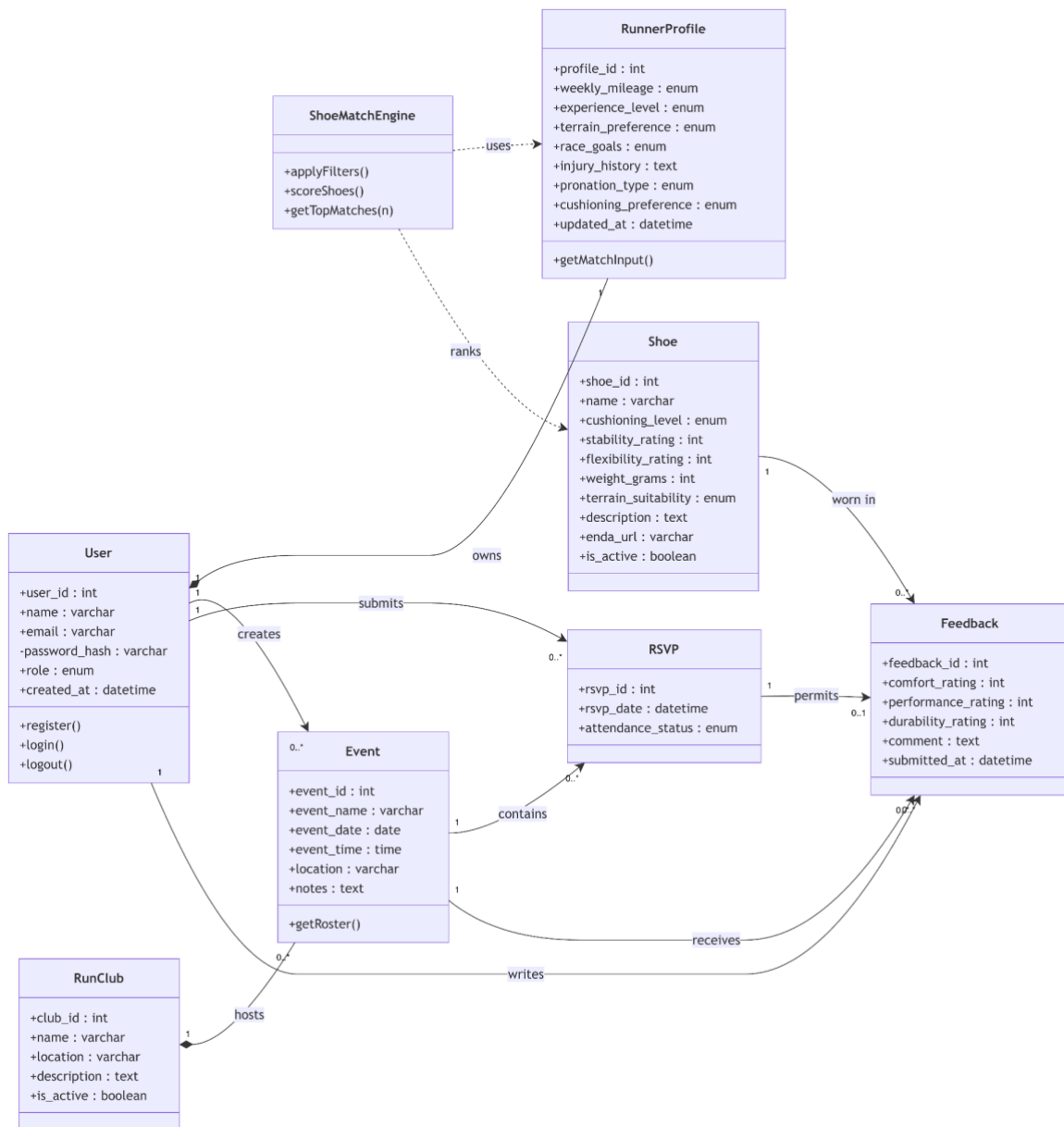
Key cardinalities: users 1–1 runner_profiles; run_clubs 1–many events; events many–many users (via rsvps); rsvps 1–0..1 feedback; shoes 1–many feedback.



Design Class Diagram

The design class diagram reflects the object-oriented structure of the platform's PHP application layer across eight classes — seven mapping directly to database entities, plus the ShoeMatchEngine, which encapsulates the shoe matching business logic and is not persisted to the database.

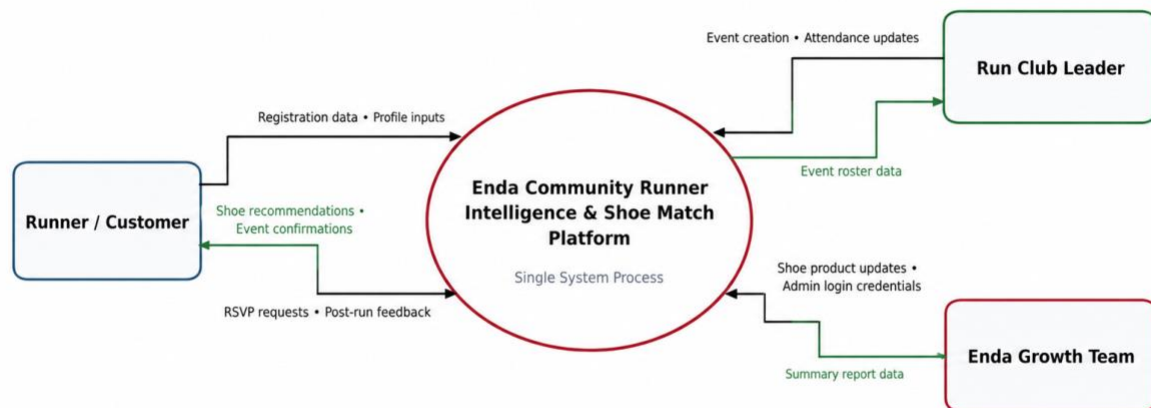
The central class is **User**, which composes a one-to-one **RunnerProfile**. The **ShoeMatchEngine** takes a **RunnerProfile** and the active **Shoe** catalog as inputs, applying filters and weighted scoring to return ranked recommendations. **RunClub** composes one-to-many **Events**; each Event aggregates **RSVP** records (resolving the many-to-many between User and Event), and each RSVP optionally permits one **Feedback** submission.



Context Diagram (System-Level DFD)

The context diagram presents the Enda platform as a single system bubble surrounded by its three external actors. The Runner/Customer sends registration data, profile inputs, RSVP requests, and post-run feedback into the system, and receives shoe recommendations and event confirmations in return. The Run Club Leader sends event creation and attendance update inputs, and receives event roster data. The Enda Growth Team sends shoe product updates and admin login credentials, and receives summary report data.

The context diagram establishes the full scope of external data flows at the highest level of abstraction before decomposition into the Level 0 DFD.\

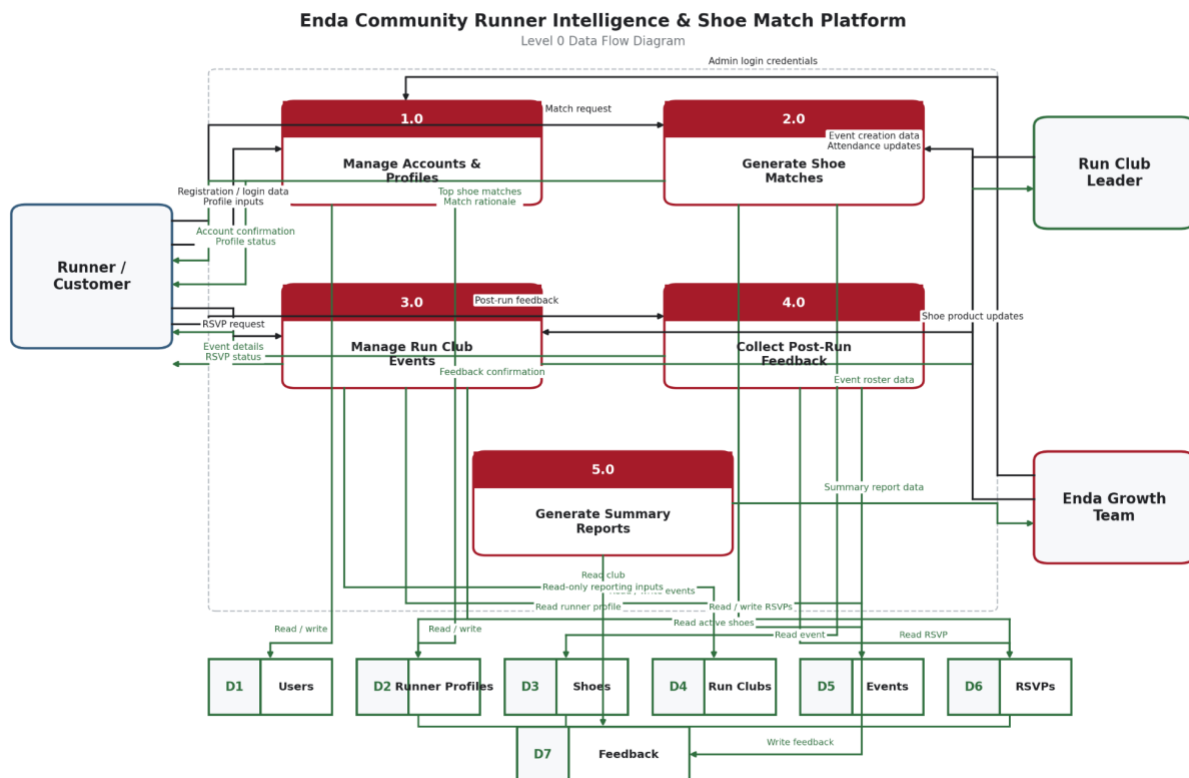


Context-level rule: Internal processes and data stores are intentionally omitted.

Level 0 Data Flow Diagram

The Level 0 DFD decomposes the system into five major processes and seven data stores, showing the primary data flows between external actors and the system boundary.

Process	Reads From	Writes To	External Actors
1.0 Manage Accounts & Profiles	D1 (Users), D2 (Runner Profiles)	D1, D2	Runner (input/output), All users (login)
2.0 Generate Shoe Matches	D2 (Runner Profiles), D3 (Shoes)	— (dynamic, no write)	Runner (receives results)
3.0 Manage Run Club Events	D4 (Run Clubs), D5 (Events), D6 (RSVPs)	D5, D6	Leader (creates/manages), Runner (RSVPs)
4.0 Collect Post-Run Feedback	D5 (Events), D6 (RSVPs), D7 (Feedback)	D7	Runner (submits feedback)
5.0 Generate Summary Reports	D2, D3, D5, D6, D7	— (read-only)	Enda Growth Team (views reports)



Level 1 DFD — Process 2.0: Generate Shoe Matches

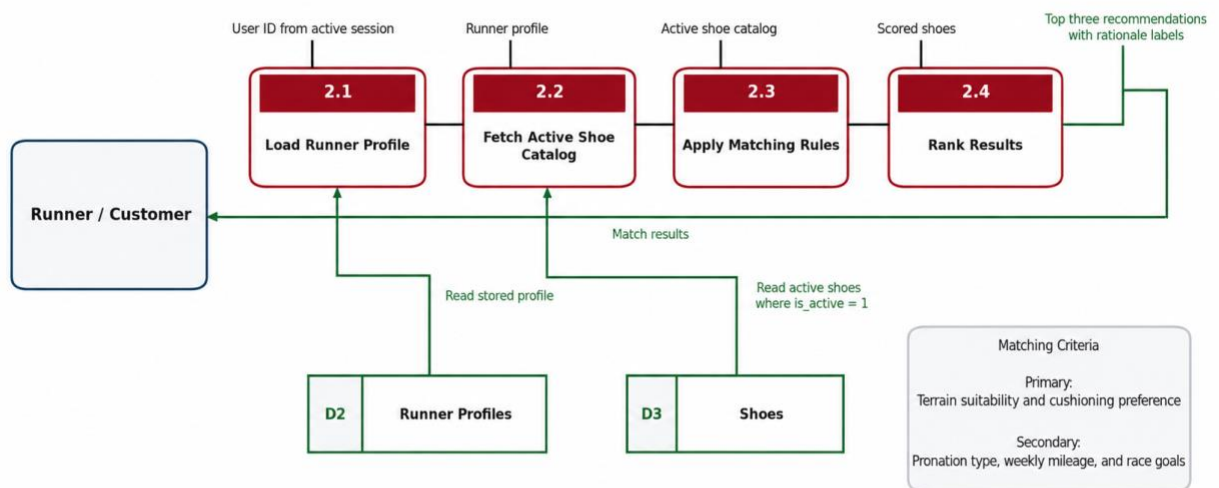
This diagram zooms into Process 2.0 and decomposes the shoe match engine into five sub-processes:

2.1 Load Runner Profile — retrieves the runner's stored profile from D2 (Runner Profiles) using their user ID from the active session.

2.2 Fetch Active Shoe Catalog — queries D3 (Shoes) and returns only records where `is_active = 1`, excluding retired products.

2.3 Apply Matching Rules — scores each active shoe against the runner's profile. Primary filters: terrain suitability and cushioning preference must match. Secondary scoring: pronation type, weekly mileage, and race goals contribute weighted points to the overall score.

2.4 Rank Results — sorts the scored shoe list in descending order and selects the top three.

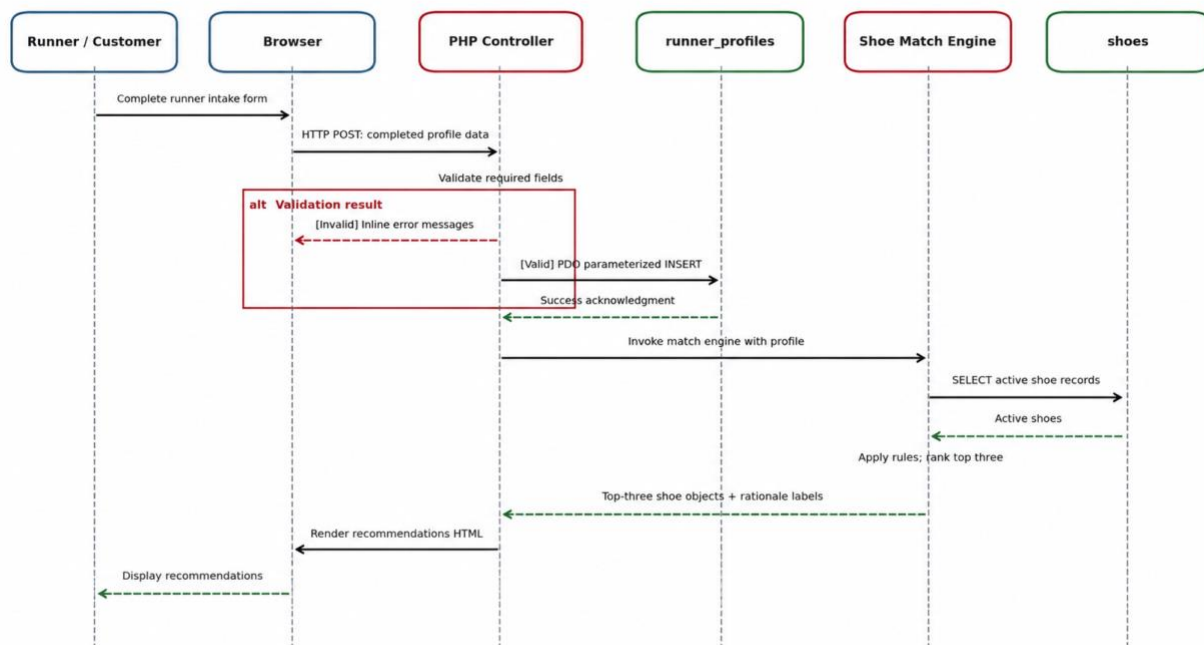


Sequence Diagram — Profile Creation and Shoe Recommendation (UC-02 / UC-03)

This diagram models the end-to-end interaction for the most critical user flow: completing the runner intake form and receiving shoe recommendations.

The sequence begins when the Browser sends an HTTP POST to the PHP Controller with the completed profile form data. The PHP Controller validates all required fields; if validation fails, it returns inline error messages to the browser without a database write. On successful validation, the controller executes a parameterized INSERT via PDO into the runner_profiles table and receives a success acknowledgment.

The controller then invokes the Shoe Match Engine, which queries the shoes table for active records, applies matching logic, and returns an ordered array of the top three shoe objects with rationale labels. The controller writes no additional data at this stage — the match results are computed dynamically on each profile load. Finally, the controller renders the recommended HTML page and returns it to the browser.



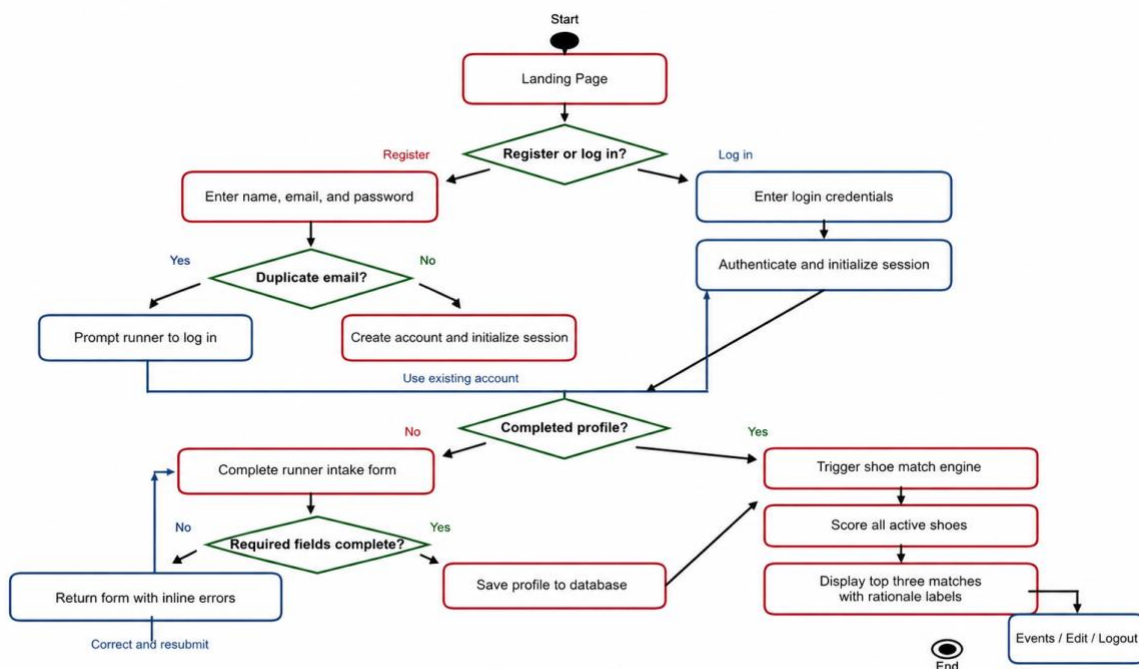
Activity Diagram — Runner Onboarding Flow

This diagram models the runner's complete onboarding path from first visit through receiving shoe recommendations, capturing the decision points and alternate paths in the flow.

The activity begins at the Landing Page, where the runner chooses to register or log in. On the registration path: the runner enters their name, email, and password → the system checks for a duplicate email → if a duplicate exists, the runner is prompted to log in instead → if unique, the account is created and a session is initialized.

Following account creation (or login with an existing account), the system checks whether the runner has a completed profile. If no profile exists, the runner is directed to the intake form. The runner fills out all required fields (mileage, terrain, experience, goals, injury history, pronation, cushioning preference) and submits. The system validates the form → if incomplete, the form is returned with inline errors → if complete, the profile is saved to the database.

The match engine is triggered automatically on profile save, scores all active shoes, and the runner is redirected to the Shoe Recommendations page displaying their top three matches with rationale labels. From this point the runner may navigate to event listings, edit their profile, or log out.



Appendix G: Source Code

The full source code for the Enda Community Runner Intelligence and Shoe Match Platform is maintained in the team's GitHub repository. A zip file containing the complete codebase — including all PHP controllers, HTML templates, CSS, JavaScript, and the MySQL schema dump — is submitted alongside this report as a separate deliverable attachment.

[Source code zip file: Enda_Platform_SourceCode.zip — attached]

The codebase is organized as follows:

- /index.php — Landing page
- /register.php, /login.php — Authentication
- /profile.php — Runner profile form (UC-02, UC-10)
- /match.php — Shoe match engine logic
- /recommendations.php — Shoe recommendations view (UC-03)
- /events.php, /event_detail.php — Events listing and RSVP (UC-04)
- /feedback.php — Post-run feedback form (UC-05)
- /admin/events.php, /admin/roster.php — Leader event management (UC-06, UC-07)
- /admin/products.php — Shoe product management (UC-11)
- /admin/reports.php — Summary reports dashboard (UC-08)
- /db/schema.sql — MySQL database schema
- /db/seed_shoes.sql — Enda shoe product seed data